Optimal Solar Panel Battery Management at the Household Scale

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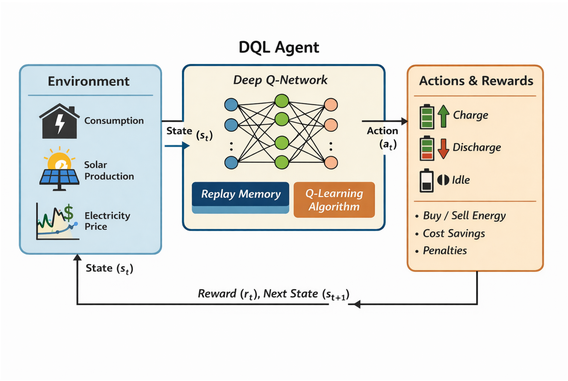
Objectives

As solar panels and home batteries become more widely available, the question of how to best manage energy storage is becoming increasingly important. A household equipped with these systems must constantly decide: should it charge the battery or feed its excess production back into the grid? The challenge stems from uncertainty regarding consumption, solar production, and electricity prices. We propose a realistic simulation environment coupled with a reinforcement learning algorithm to automatically learn an optimal management strategy.

Introduction

The rapid deployment of residential photovoltaic systems has transformed households into small-scale energy producers. In the past, surplus electricity could be resold to EDF at a fixed regulated tariff, providing predictable revenues and reducing the need for complex optimisation strategies. Today, feed-in tariffs are less advantageous or more regulated, and households increasingly rely on self-consumption and battery storage to maximise the value of their production. At the same time, solar production remains variable and electricity prices can fluctuate due to time-of-use or dynamic market mechanisms. These factors make optimal battery management a non-trivial sequential decision problem.



Environement

The environment action space is composed of 3 actions : charging the battery, discharging it, or stay idle. The observation space is composed of the battery current charge, the current time step, and a forecast (from current time to x steps ahead) of the house electric consumption, the house solar production, and electricity price. The agent is rewarded proportionnaly to the quantity of electricity it buys or sells, depending on the quantity needed or in excess (production - consumption). Consumption, production and price forecasts are updated in a first-in first-out fashion according to the models described in the following section. It is worth noting that the battery is charged and discharged in a single time step.

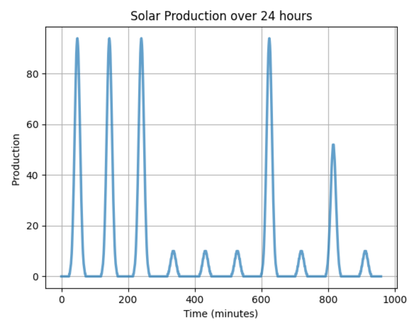
Models

Goal: Realistic stochastic environment for control.

Consumption. Mean-reverting dynamics (Ornstein–Uhlenbeck). Parameters μ and ν fitted from a real household data .

$$dL(t) = \theta(\mu(t) - L(t))dt + \sigma(t)dW(t) \qquad (1)$$

Solar production. Gaussian profile centered at midday. Zero at night. Scaled by weather ($e_s \in \{0.1, 0.5, 0.9\}$).



Electricity price. Either fixed tariff (peak/off-peak) or stochastic process.

Deep Q Learning

The control policy is optimized using Deep Q Learning, which consists in approximating the action-value function $q : (\mathcal{S}, \mathcal{A}) \mapsto \mathbb{R}$ with a neural network. The model is trained on batches of action-state observations stored in a replay memory that is updated with each steps, To stabilize training, two networks are maintained: a policy network updated at every step, and a target network updated softly via $\theta' \leftarrow \tau\theta + (1-\tau)\theta'$. The policy follows an ε-greedy strategy, decaying from full exploration to near-greedy exploitation over training. The network is optimized by minimizing the Huber loss between predicted q-values and the Bellman targets:

$$\mathcal{L} = \mathbb{E}\left[\mathcal{H}_\delta(r + \gamma \max_{a'} q_{\theta'}(s', a') - q_\theta(s, a))\right]$$

Results

We ran experiments with 500 episodes, using a random policy then using the DQL policy estimation. We see that the random policy yield an average reward of about -0.4 cf fig 1, whereas the DQL agent converges as soon as 100 episodes to a reward of about -0.05 cf fig 2. On figure 3 we can see the actions taken by the DQL agent during the last 3 episodes : the agent is favoring charging and discharging actions instead of being idle.

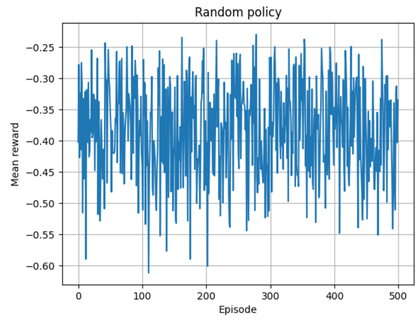


Figure 1:Mean reward on 500 episodes with random policy

This can be partly explained by the fact that the reward for being idle and discharging the battery is quite similar if the battery charge is low, and that the charge and discharge is instant.

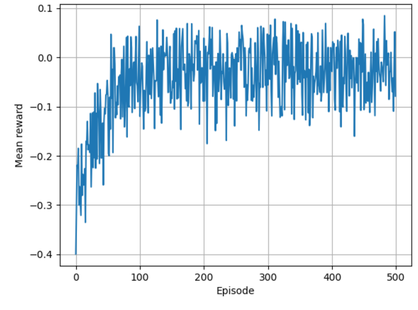


Figure 2:Mean reward on 500 episodes with DQL

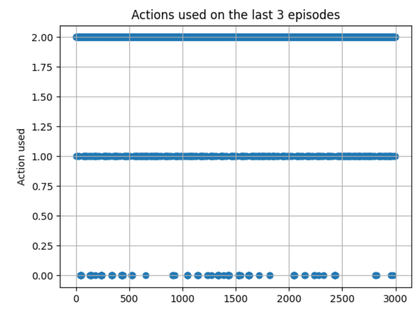


Figure 3:Actions used on the last 3 episodes (0:idle, 1:charge, 2:idle

References

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Reinforcement Learning (DQN) Tutorial.
Technical report, PyTorch, 2017, update 2025.

[2] UCI Machine Learning Repository.
Individual household electric power consumption dataset, 2012.

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